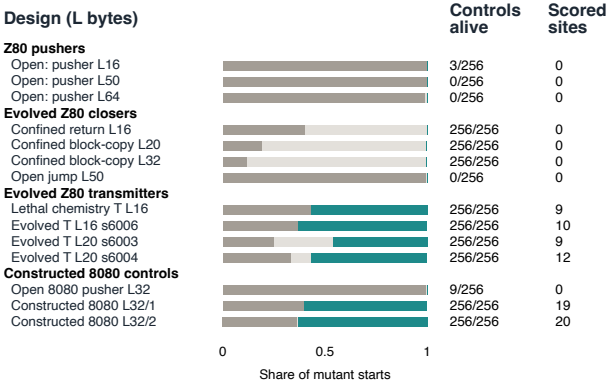
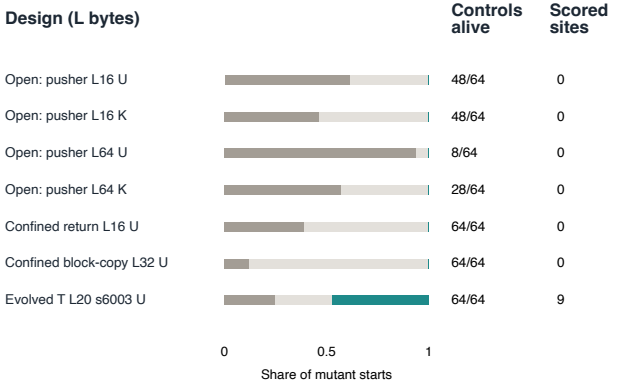
a All historical serial designs: transfer 8



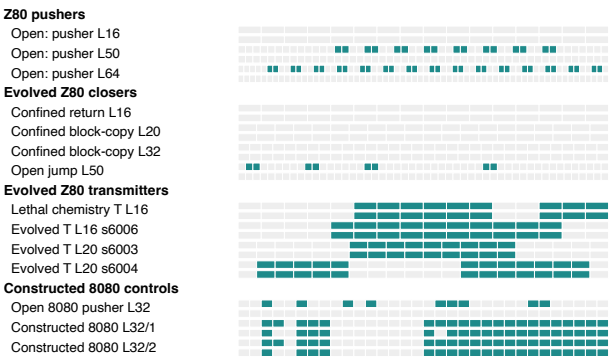
Dark grey lost; light grey alive noncarrier; teal alive carrier
Carrier = tape contains centred cyclic mutant 5-byte window
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess = carriers/survivors (mutant) – carriers/survivors (WT)

b Branching generation-8 outcomes: U / K



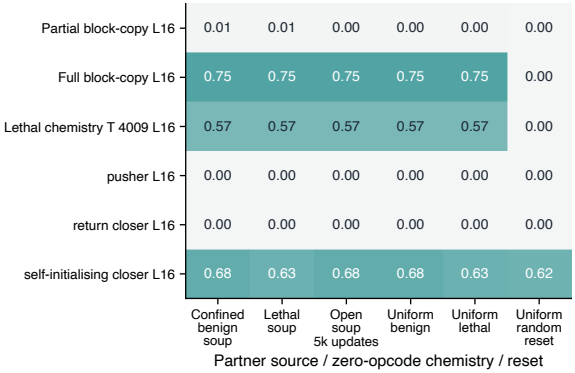
Grey lost; pale alive <50% carriage; teal alive ≥50% carriage
Carriage = members containing mutant 5-byte window / all members
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess as a; U uniform/K soup; 32 values×4 starts/site; WT 64

c Complete serial site maps: 1 / 8



Upper: serial transfer 1; lower: serial transfer 8; teal = scored site
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Same excess definition as a; tracks span bytes 0 to L−1

d All partner-environment carriage tests



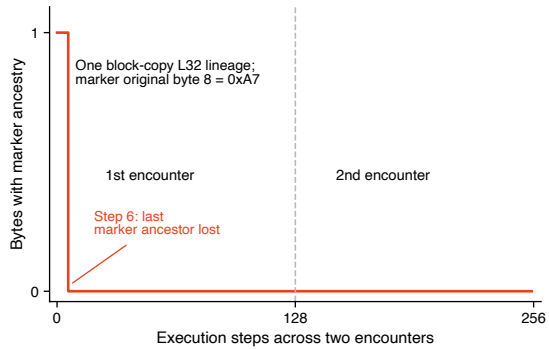
Generation-8 carriers / all mutant starts (serial definition in a)
All 6 parents L16: 16×255×16 = 65,280 starts/design; WT 256
Benign zero executes; lethal zero halts; open soup at 5,000 updates

e Identifiable alternatives and controls

Design	Identifiable / tested alternatives	WT starts
Z80 pushers		
Open: pusher L16	4080/4080	256
Open: pusher L50	1600/1600	256
Open: pusher L64	2048/2048	256
Evolved Z80 closers		
Confined return L16	4080/4080	256
Confined block-copy L20	5100/5100	256
Confined block-copy L32	8160/8160	256
Open jump L50	1600/1600	256
Evolved Z80 transmitters		
Lethal chemistry T L16	4080/4080	256
Evolved T L16 s6006	4080/4080	256
Evolved T L20 s6003	5100/5100	256
Evolved T L20 s6004	5100/5100	256
Constructed 8080 controls		
Open 8080 pusher L32	8160/8160	256
Constructed 8080 L32/1	8160/8160	256
Constructed 8080 L32/2	8160/8160	256

Identifiable: marker window absent from WT; T = evolved transmitter
Alternative types = L byte positions × tested values
16 starts/value; pusher50/64 & jump50: 32 values, others 255

f Ordered ancestry: two full encounters



Counted unit: bytes with byte-8 ancestry; all-zero byte partners
Step 6 reads address 4 (ancestry byte 0), writes 0x04 over 8
At execution step 128: offspring becomes parent; processor reset